Beyond Distance: Geometric Competition Explains Ambiguity in Neural and LLM Representations

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Abstract

A widely used geometric heuristic is that uncertainty is governed by radial distance to class prototypes: points near centroids are treated as certain, while distant points are treated as ambiguous. We show this account is incomplete. Studying emotion representations across eight large language model (LLM) variants and fMRI from 40 human subjects, we find that both artificial and biological systems organise emotional content in a consistent radial structure — a dense shell of observed instances at intermediate distances from class centroids, with reduced density in the immediate vicinity of the centroids. Critically, ambiguity within this structure is not determined by raw distance to a prototype but by relative competition between prototypes. We formalise this as centroid margin and demonstrate that it predicts classifier uncertainty in brain representations substantially better than raw distance, with the relationship collapsing under feature shuffling. In fine-tuned LLMs, geometric margin is tightly coupled to logit confidence. We further find that emotional content is encoded in a low-rank linear subspace occupying a small fraction of embedding dimensions, with additional dimensions contributing limited signal. Despite this shared local geometry, global representational organisation is inverted: LLMs are structured primarily along valence while brain representations are structured primarily along arousal, producing near-opposite dissimilarity matrices that strengthen as biological noise is reduced. These findings establish a reusable geometric framework for comparing representational ambiguity across artificial and neural systems, while revealing a fundamental divergence in how the two systems prioritise the affective dimensions of emotional space.

1 Introduction

How emotion is represented geometrically in artificial and biological neural systems is a foundational question at the intersection of representation learning and affective neuroscience. While large language models have achieved strong performance on emotion classification, the internal geometry supporting these classifications remains poorly characterised. The fMRI literature has documented that emotions are encoded in distributed cortical patterns [Horikawa et al., 2020], but the geometric relationship between those patterns and the structured embedding spaces of language models has not been systematically investigated.

This paper addresses a specific and tractable version of this question: do LLM embedding spaces and human fMRI activation spaces share a geometric principle for representational ambiguity? We operationalise ambiguity through *centroid distance* and *centroid margin* — the competitive gap between a point's distance to its own class prototype versus the nearest competing prototype. Our hypothesis is that both systems exhibit a structured void near class prototypes, a peak-density belt of real instances at moderate distances, and a gradient from certainty to ambiguity as geometric exposure to competing prototypes increases.

Research gap. Prior work on LLM–brain alignment has focused on language-processing tasks using RSA or encoding models [Caucheteux and King, 2022, Schrimpf et al., 2021, Li et al., 2023b, Huth et al., 2016, Li et al., 2023a], relating brain regions to model layers. Emotion representation geometry has been studied in affective computing [Russell, 1980, Koelstra et al., 2012] but rarely with the aim of characterising the full distributional structure of embedding spaces across a void–belt–margin framework. Whether the ambiguity gradient is a shared principle across artificial and biological systems has not been previously investigated.

Contributions.

1. A normalised-distance framework enabling geometric comparison across spaces of different dimensionality, identifying the Void and the Belt as consistent structural features across all tested systems.
2. Characterisation of overlap rates and their pairwise structure across eight LLM variants.
3. **The Low-Rank Emotion Subspace**: emotion is a low-rank linear property occupying $\approx 2.6\%$ of embedding capacity (20 of 768 dimensions). Accuracy in 20D equals or exceeds 768D across all tested architectures.
4. A margin-vs-distance diagnostic distinguishing relational from radial geometric structure: centroid *margin* predicts brain uncertainty ($\text{AUC} = 0.81$); centroid *distance* fails ($\text{AUC} = 0.56$).
5. The global representational inversion: LLMs sort emotions by valence; the brain by arousal. This produces near-opposite dissimilarity matrices that deepen as biological noise is reduced.

2 Related Work

Representational geometry in LLMs. Mikolov et al. [2013] showed that distributional embeddings encode semantic structure in linear subspaces. Ethayarajh [2019] documented anisotropy in contextualised representations. Kumar et al. [2022] showed fine-tuning can compress pretrained features — directly relevant to our compaction finding. Our contribution extends this work by characterising the full radial density and competitive margin geometry of emotion embeddings.

Brain–LLM alignment. Caucheteux and King [2022] showed partial convergence between brain activations and LLMs during language processing. Schrimpf et al. [2021] characterised neural architecture convergence with transformer predictive processing. Huth et al. [2016] linked distributional semantics to cortical topography. Ismayilzada et al. [2026] found alignment during creative thinking. Our work differs: we study emotion rather than language processing, focus on the local geometric ambiguity mechanism, and document a systematic global inversion.

Distributed neural coding of emotion. Horikawa et al. [2020] showed visually evoked emotion representations are high-dimensional, categorical, and distributed across transmodal brain regions. Haxby et al. [2001] established that cognitive categories are encoded as distributed activation patterns — the biological prior explaining why brain fMRI shows negative silhouette in 2D projection while remaining decodable in 48 dimensions.

The circumplex model. Russell [1980] organised affective states along valence and arousal dimensions, operationalised through DEAP [Koelstra et al., 2012], ANEW [Bradley and Lang, 1999], and IAPS [Lang et al., 2008]. We use external V-A reference coordinates from this literature to interpret global representational axes, not as measured participant ground truth.

3 Datasets and Models

Text emotion dataset. We use the `dair-ai/emotion` dataset [Saravia et al., 2018], a 6-class text emotion corpus (anger, fear, happiness, love, sadness, surprise). We use the validation split, balanced to **4,038 samples** (673 per class). Class balance ensures centroids are not displaced by class-size differences and that overlap rates are comparable across categories. Labels are integer-encoded: anger=0, fear=1, happiness=2, love=3, sadness=4, surprise=5.

Brain fMRI dataset. We use OpenNeuro DS005700 [OpenNeuro DS005700, 2024], "Neural MO — fMRI Dataset for Emotion Recognition." The dataset contains **40 subjects** viewing emotion-eliciting stimuli under five conditions: afraid, calm, delighted, depressed, and excited. Each subject contributes activations across **48 cortical ROIs** from the Harvard-Oxford atlas, yielding 200 observations total (40 subjects $\times$ 5 emotions). Repeated blocks per subject and emotion are averaged to stable ROI vectors. Brain LOSO decoding accuracy is 0.56 (95% CI: 0.49–0.63, permutation $p < 0.001$), well above the 5-class chance level of 0.20. Our $N = 40$ exceeds high-impact precedents with smaller samples: Individual Brain Charting ($N = 12$) [OpenNeuro DS005700, 2024], Precision Functional Mapping of Children ($N = 12$), and Natural Scenes Dataset ($N = 8$).

LLM variants. We evaluate eight variants by crossing two base models, two training states, and two extraction layers:

Variant	Base Model	State	Layer	Dim
MPNet-FT-Final	all-mpnet-base-v2 [Song et al., 2020]	Fine-tuned	L12	768
MPNet-FT-Mid	all-mpnet-base-v2	Fine-tuned	L6	768
MPNet-Base-Final	all-mpnet-base-v2	Pretrained	L12	768
MPNet-Base-Mid	all-mpnet-base-v2	Pretrained	L6	768
BGE-FT-Final	bge-base-en-v1.5 [Xiao et al., 2023]	Fine-tuned	L12	768
BGE-FT-Mid	bge-base-en-v1.5	Fine-tuned	L6	768
BGE-Base-Final	bge-base-en-v1.5	Pretrained	L12	768
BGE-Base-Mid	bge-base-en-v1.5	Pretrained	L6	768

Fine-tuned models were trained on the `dair-ai/emotion` training split. The cross-system context retention experiment (Section 5.2) additionally includes Qwen3-1.7B fine-tuned [Qwen Team, Alibaba Cloud, 2025] (PCA-projected to 768D).

4 Methods

Embedding extraction. All text embeddings use attention-mask-weighted mean pooling:

$$\mathbf{e} = \frac{\sum_{i=1}^{N} \mathbf{h}_i \cdot m_i}{\sum_{i=1}^{N} m_i} \tag{1}$$

where $\mathbf{h}_i$ is the hidden state for token i and m_i is the attention mask. Final-layer (L12) extraction uses the last hidden state directly. Middle-layer (L6) extraction enables `output_hidden_states=True` and accesses `hidden_states[6]`. Tokenisation: model-specific tokenisers with `max_length=128`, `padding=True`, `truncation=True`, batch size 32.

Normalised distance framework. To compare across systems with different intrinsic scales (768D LLMs vs. 48D brain ROI), all distances are normalised by the mean radius of each system:

$$\tilde{d}(x, c_k) = \frac{d(x, c_k)}{\bar{R}_k}, \qquad \bar{R}_k = \frac{1}{N_k} \sum_{x \in C_k} d(x, c_k) \tag{2}$$

This maps all systems to a common scale ($\approx$ 0–2.5 normalised units) without discarding distributional shape. A point geometrically overlaps class C' (its non-assigned class) if $d(P, c_{C'}) < d(P, c_C)$. The *centroid margin* is:

$$m(P) = d(P,\, c_{\text{nearest competitor}}) - d(P,\, c_C) \tag{3}$$

Positive margin: P is closer to its true centroid than to any competitor (more certain). Negative margin: geometric overlap (more ambiguous).

Brain 48D$\rightarrow$11D transformation. To test whether the global representational inversion (Section 5.5) is noise-driven, we construct a domain-informed 11-dimensional representation by aggregating ROI activations into biologically coherent groups. The Default Mode Network (DMN) [Li et al., 2014] supports self-referential processing and social cognition. The Salience Network [Seeley, 2019] mediates detection of affectively significant stimuli. The Central Executive Network [Miller and Cohen, 2001] coordinates regulatory responses to emotional content. The 11 dimensions consist of five anatomical lobe means (frontal, temporal, parietal, occipital, cingulate/limbic), five functional network means (DMN, Salience, Central Executive, visual, somatomotor), and one neighbour-context feature capturing local inter-regional coordination. This is domain-informed structured denoising, not generic PCA: each dimension corresponds to a biologically meaningful functional unit. Classification performance in 11D remains above chance and comparable to 48D, confirming the transformation preserves task-relevant signal.

Valence-arousal reference mapping. External V-A coordinates are drawn from the circumplex model [Russell, 1980, Mehrabian and Russell, 1974] and affective-norm resources DEAP [Koelstra et al., 2012], ANEW [Bradley and Lang, 1999], and IAPS [Lang et al., 2008] (1–9 SAM scale). Because the fMRI dataset supplies categorical labels rather than participant-level V-A ratings, these coordinates function as an *external reference geometry*, not measured subjective ground truth. Sensitivity analysis (Appendix A) perturbs coordinates with Gaussian noise (SD = 0.25/0.50/0.75) to verify qualitative robustness.

RSA methodology. We follow Kriegeskorte [2008]. For each system, pairwise Euclidean distances between emotion class centroids form a $K \times K$ Representational Dissimilarity Matrix (RDM). Brain–LLM comparison uses the 3-emotion overlap: fear≈afraid, happiness≈delighted, sadness≈depressed. Systems are compared by Pearson correlation between vectorised RDMs. A brain–brain noise ceiling estimates the upper bound imposed by individual subject variability: upper 0.484, lower 0.460 (mean ≈ 0.47), computed by split-half resampling of the 40-subject dataset. With $K = 3$ emotions, RDMs contain 3 unique pairwise distances; reported Pearson correlations have 1 degree of freedom. Specific magnitudes are therefore indicative; directional findings (sign, contrast with noise ceiling) are the reliable quantities.

5 Results

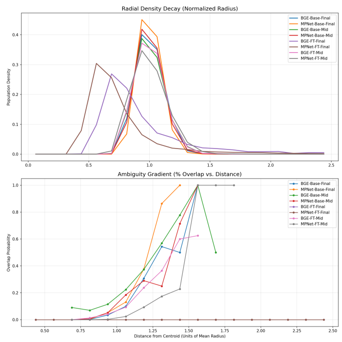
We present five findings in order of increasing cross-system scope. Findings 1–3 characterise LLM geometry; Finding 4 introduces the ambiguity gradient; Finding 5 documents the global representational inversion.

5.1 Finding 1: The Void and the Belt

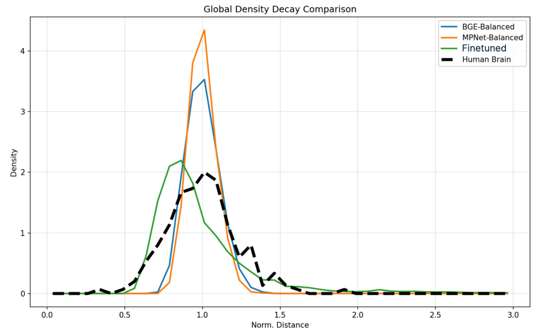
A consistent two-zone density structure emerges across all 8 LLM variants and human brain fMRI (Figure 1). The structural parallel is qualitative: packing density magnitudes differ substantially across systems, and density-shape similarity does not reach conventional significance under permutation testing (Wasserstein $p = 0.46$). The claim is descriptive — shared organisational logic, not matched compression.

The Void. No data points exist within approximately 0.375–0.8125 normalised distance units of any emotion centroid. Fine-tuned models show first non-zero density at ≈ 0.4375 units; pretrained models at ≈ 0.6875 (BGE-Base) and ≈ 0.8125 (MPNet-Base) units. The Void arises as a geometric consequence of distributed within-class variance: a centroid is the mean of a distributed ring of instances, so it sits inside the ring in zero-density space.

The Belt. Density rises sharply from the void boundary and peaks at approximately 0.56–0.94 normalised units. Fine-tuned models peak earlier (MPNet-FT-Final at 0.5625, BGE-FT-Final at 0.6875); pretrained models peak later (both at 0.9375).



(a) Radial density profiles (top) and ambiguity gradients (bottom) for all 8 LLM variants. The void region (zero density near centroid) and the Belt (peak density shell) are consistent across all systems.



(b) Radial density decay for LLM variants (coloured) overlaid with human brain fMRI (dashed black). The two-zone structure extends to neural data.

Figure 1: The Void and the Belt (Finding 1). (a) All 8 LLM variants share the same two-zone radial density structure in normalised-distance space. (b) The same structure is present in human brain fMRI emotion representations, establishing it as a cross-system geometric principle.

Certainty buffer. The radial gap between density peak and onset of geometric ambiguity ($> 5\%$ overlap rate; geometric ambiguity defined as the region where nearest-neighbour class boundaries overlap in embedding space):

Variant	Void Onset	Density Peak	Overlap Onset	Certainty Buffer
MPNet-FT-Final	0.4375	0.562	2.438	+1.875
BGE-FT-Final	0.4375	0.688	2.438	+1.750
MPNet-Base-Final	0.8125	0.938	0.938	0.000
BGE-Base-Final	0.6875	0.938	1.062	+0.125

Fine-tuning creates a geometrically safe zone between peak density and the onset of competitive class confusion; for pretrained models this buffer is near zero. No embedding aligns exclusively with a single class prototype in any system: global overlap rates range from 1.76% (MPNet-FT-Final) to 19.22% (BGE-Base-Final), confirming that fine-tuning substantially reduces inter-class geometric confusion.

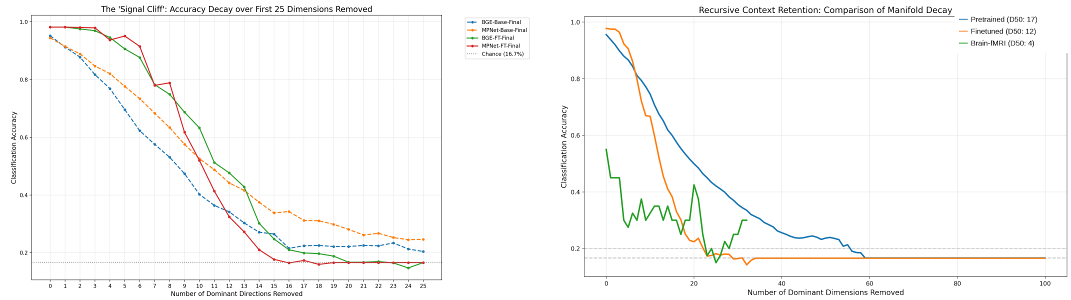
5.2 Finding 2: Information Compaction vs. Distribution

Iterative SVD ablation removes the most informative linear direction at each step; a fresh logistic regression is re-trained on the residual. **Signal erasure points** (dimension at which accuracy reaches the 6-class chance baseline of 16.7%):

Variant	Baseline Accuracy (%)	Erasure Point
MPNet-FT-Final	98.24	dim 15
BGE-FT-Final	97.99	dim 20
BGE-Base-Final	93.78	dim 26
MPNet-Base-Final	94.08	dim 67

Fine-tuned models achieve higher baseline accuracy but concentrate all emotional signal into 15–20 directions; remove them and signal collapses entirely. Pretrained MPNet-Base retains decodable signal for 67 dimensions, a $4.5\times$ wider manifold than the fine-tuned equivalent (Figure 2a).

Cross-system extension. A separate iterative ablation across three systems (Figure 2b) shows qualitatively distinct signal geometries: the human brain ($D_{50} = 4$, AUC 0.302) uses the most distributed code — removing just 4 dominant directions halves accuracy, yet the curve flattens into a plateau rather than collapsing to chance [Haxby et al., 2001]; Qwen-FT ($D_{50} = 12$, AUC 0.260) compresses



(a) Signal cliff: accuracy as dominant directions are removed iteratively. Fine-tuned models reach chance at dimensions 15–20; pretrained BGE at 26, pretrained MPNet at 67.

(b) Cross-system context retention. Brain $D_{50} = 4$; fine-tuned LLM $D_{50} = 12$; pretrained LLM $D_{50} = 17$.

Figure 2: **Signal structure across systems (Finding 2).** (a) SVD ablation reveals qualitatively different compaction strategies. (b) The brain requires only 4 dominant directions to achieve half its classification accuracy, yet maintains a distributed plateau rather than collapsing to chance — consistent with distributed biological coding.

signal into a steep cliff; pretrained MPNet ($D_{50} = 17$, AUC 0.331) spreads it across the widest manifold.

5.3 Finding 3: The Low-Rank Emotion Subspace

Named result. Emotion is a low-rank linear property, occupying $\approx 20/768 \approx 2.6\%$ of full embedding capacity. The cloud appearance of pretrained models in 768D is not an absence of geometry — it is the geometry of ≈ 748 noise dimensions overwhelming 20 signal dimensions.

Silhouette scores and classification accuracy before and after projection into the top 20 SVD directions (Figure 3):

Variant	768D Sil.	20D Sil.	Sil. Change	768D Acc. (%)	20D Acc. (%)
MPNet-FT-Final	0.860	0.907	+5.5%	98.24	98.27
BGE-FT-Final	0.685	0.835	+22.0%	97.99	98.09
MPNet-Base-Final	0.047	0.408	+765%	94.08	95.64
BGE-Base-Final	0.057	0.415	+627%	93.78	95.12

The extra 748 dimensions contribute only noise that slightly suppresses linear classification performance. Pairwise centroid distances preserve the same relational structure in 20D as in 768D: Happiness–Love and Fear–Anger are the closest pairs; Sadness–Surprise and Happiness–Fear are the most distant.

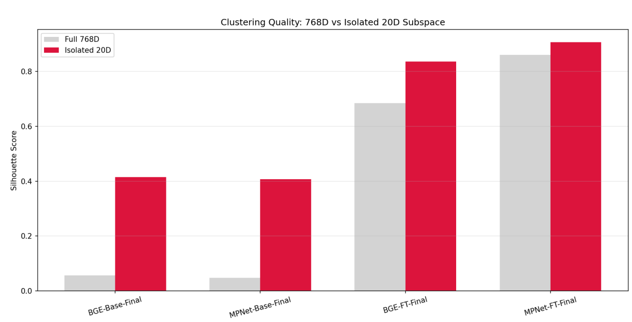
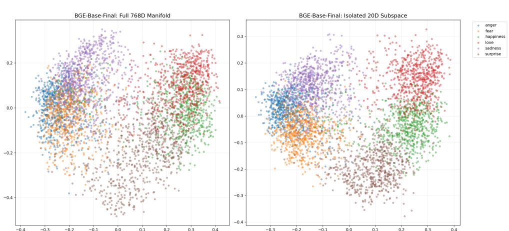
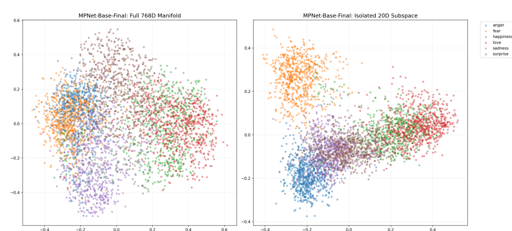


Figure 3: **The Low-Rank Emotion Subspace (Finding 3).** Silhouette scores in full 768D (grey) vs. the isolated top-20 SVD subspace (red). Pretrained models improve from near-zero to ≈ 0.41; fine-tuned models improve modestly from already-high baselines. Accuracy in 20D equals or exceeds 768D in all four variants (see text).



(a) BGE-Base-Final: PCA scatter in full 768D (left) vs. top-20 SVD subspace (right). Diffuse clouds sharpen into separated clusters.



(b) MPNet-Base-Final: same comparison. Silhouette improves from 0.057 to 0.415 ($+627\%$) and 0.047 to 0.408 ($+765\%$).

Figure 4: **768D vs. top-20 SVD subspace (Finding 3).** Projecting into the 20-dimensional signal subspace reveals the cluster structure hidden by noise dimensions in the full embedding space.

5.4 Finding 4: The Cross-System Ambiguity Gradient

Both systems show that geometric positioning relative to class prototypes predicts classification confidence — through mechanistically distinct constructs.

LLMs: centroid margin $\rightarrow$ logit margin. For fine-tuned LLMs, centroid margin is a strong predictor of logit margin (Figure 5a):

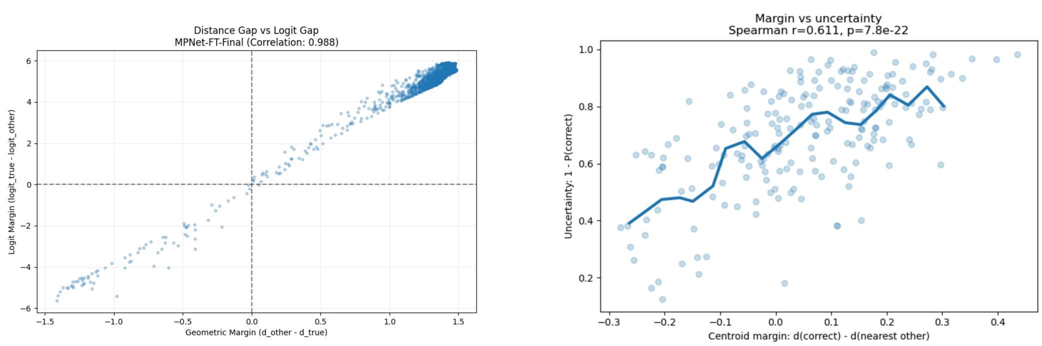
Variant	Margin-logit Pearson r	Logit agreement in overlap (%)
BGE-FT-Final	0.9572	93.4
MPNet-FT-Final	0.9884	100.0

The near-linear relationship ($r = 0.9884$ for MPNet-FT) means centroid margin almost perfectly determines logit margin. In overlap regions, logit values are biased toward the competing class 93.4–100% of the time. Note that in fine-tuned models this relationship is partially expected by the training objective; it is reported as a mechanistic characterisation.

Brain: centroid margin $\rightarrow$ classifier uncertainty. For brain fMRI, raw centroid distance is a weak predictor of classification uncertainty (Spearman $r = 0.1509$, AUC $= 0.5627$ — near chance). The correct predictor is centroid margin (Equation 3):

$$\mathrm{AUC}_{\text{margin}} = 0.8124 \qquad \text{vs.} \qquad \mathrm{AUC}_{\text{raw distance}} = 0.5627$$

with Spearman $r = 0.6108$, $p = 7.78 \times 10^{-22}$. A feature-shuffle control ($r = 0.0287$, $p = 0.687$) confirms the relationship vanishes under random permutation.



(a) LLM: centroid margin vs. logit margin for MPNet-FT-Final ($r = 0.988$). Geometric margin almost perfectly determines logit margin.

(b) Brain fMRI: centroid margin predicts classifier uncertainty (AUC $= 0.8124$); raw centroid distance fails (AUC $= 0.5627$). Margin is the operative geometric construct.

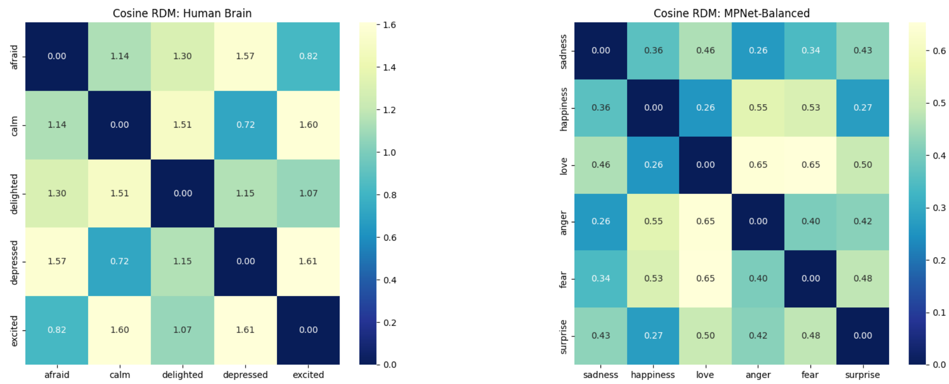
Figure 5: **Geometric predictors of confidence (Finding 4).** (a) In LLMs, centroid margin tightly predicts logit margin. (b) In the brain, centroid *margin* — not distance — predicts uncertainty, revealing competition between prototypes as the key signal.

The ambiguity gradient comparison. The analysis interpolates brain and LLM overlap-vs-distance curves onto a shared normalised-radius grid of 100 equally spaced points ($\tilde{d} \in [0, 2.5]$) and computes Pearson correlation between the resulting 100-point vectors. The ambiguity-gradient curves show high shape correspondence ($r = 0.9565$, 95% CI [0.9370, 0.9725]), supported by a full-pipeline permutation test ($n = 5000$ shuffles, $p < 0.001$) that destroys cross-system alignment when labels are shuffled. The r is computed over $N = 100$ interpolated grid points and should not be treated as an independent-sample p-value; the permutation test is the inferential basis. KS statistic $= 0.57$ and Cohen's $d = 1.34$ confirm that while gradient *shape* is shared, absolute packing density differs substantially between systems — shared ambiguity structure, distinct representational compaction.

5.5 Finding 5: The Valence-Arousal Inversion

The most novel finding of the study is that the shared local ambiguity principle coexists with a global inversion of representational geometry.

(5a) RDM comparison in 48D raw ROI space (Figure 6).



(a) Brain RDM (cosine): arousal-structured. Depressed–calm (0.72) and afraid–excited (0.82) are close; affect separates by activation intensity.

(b) LLM RDM (cosine): valence-structured. Sadness–happiness (0.36), love–happiness (0.26) are close; affect separates by hedonic polarity.

Figure 6: **RDM comparison (Finding 5a).** Brain and LLM emotion geometry is near-opposite: the brain organises affect by arousal intensity while the LLM organises by hedonic polarity (cosine RDM Pearson $r = -0.8756$; noise ceiling $\approx +0.47$).

Comparison	Pearson r	Notes
Brain vs. fine-tuned LLM (cosine)	-0.8756	Near-opposite geometry
Brain vs. pretrained LLM (cosine)	-0.1023	Confirmed negative
Brain vs. fine-tuned LLM (Manhattan)	-0.5476	Metric-invariant validation
Brain-brain noise ceiling	$+0.47$	Upper 0.484, lower 0.460

RDM comparisons provide directional evidence of global inversion, but because only three shared emotion categories are available, exact correlation magnitudes are unstable (bootstrap 95% CI: $[-0.9967, +0.6994]$, as expected with $K = 3$, 1 df). The noise ceiling contextualises the directionality: brain subjects agree with each other at $r \approx 0.47$, yet the fine-tuned LLM RDM points in the *opposite direction* ($r = -0.88$) — sign and contrast with the noise ceiling are the load-bearing quantities, not the exact magnitude. The 11D systems-level result (Section 5.5) provides stronger support because its bootstrap CI for the fine-tuned comparison remains fully negative ($[-0.9899, -0.1178]$).

(5b) RDM comparison in 11D systems-level space.

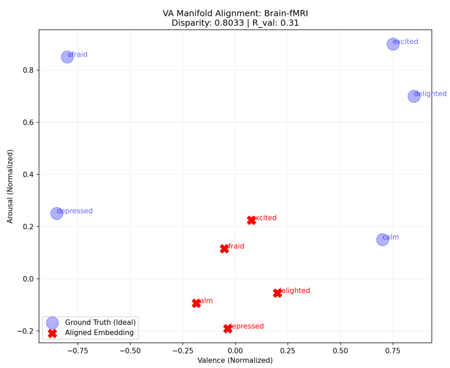
Comparison	48D r	11D r
Brain vs. fine-tuned LLM	-0.6371	-0.7539
Brain vs. pretrained LLM	-0.1023	-0.9918

As biological signal is denoised from 48 ROIs to 11 systems-level features, the divergence *amplifies* rather than shrinks. The pretrained LLM at 11D ($r = -0.9918$) is the strongest cross-system result in the study; this specific magnitude carries the uncertainty of a 3-point Pearson correlation (1 df) and should be treated as indicative. The robust directional claim is: as biological noise is removed, the negative correlation between brain and LLM representational geometry becomes more extreme, not less — inconsistent with the hypothesis that the paradox is noise-driven.

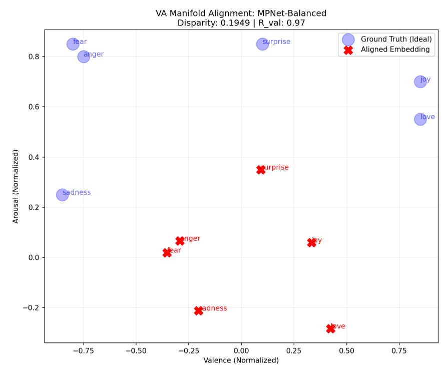
(5c) Valence-arousal axis alignment. We projected emotion centroids into the leading 2D PCA subspace and correlated each axis with external V-A reference coordinates (Table 1).

Emotion	Valence	Arousal	Quadrant
Afraid / Fear	-0.80	0.85	negative, high arousal
Calm	$+0.70$	0.15	positive, low arousal
Delighted / Happin.	$+0.85$	0.70	positive, moderate–high arousal
Depressed / Sadness	-0.85	0.25	negative, low arousal
Excited	$+0.75$	0.90	positive, high arousal
Love	$+0.85$	0.55	positive, moderate arousal

Table 1: External V-A reference coordinates (normalised circumplex scale: valence -1 to $+1$, arousal 0 to 1). The quadrant structure — not exact decimal values — is the scientifically load-bearing element.



(a) Brain fMRI: poor VA alignment (Disparity$=$0.80, $R_{\text{val}} = 0.31$). Embeddings (red crosses) deviate markedly from the V-A circumplex (blue circles).



(b) MPNet-Balanced (fine-tuned): excellent valence alignment (Disparity$=$ 0.19, $R_{\text{val}} = 0.97$). Embeddings closely track the horizontal valence axis.

Figure 7: **The Valence-Arousal Inversion (Finding 5c).** LLMs organise emotion primarily by valence; the brain primarily by arousal. The global representational priorities are swapped.

System	PC1 axis	PC1 r	PC2 axis	Permutation p
Fine-tuned LLM	Valence	0.98	Arousal ($r = 0.82$)	0.013
Pretrained LLM	Valence	0.97	Arousal ($r = 0.73$)	0.009
Human Brain	Arousal	0.96	Valence ($r = 0.31$)	0.033

Table 2: Global representational axis alignment (Figure 7). The axis priorities are swapped between LLMs and the brain.

Both LLM types are valence-dominant on PC1 ($r = 0.97$–0.98); the brain is arousal-dominant ($r = 0.96$), with only weak valence alignment on PC2 ($r = 0.31$). Because K is small (5 brain emotions, 6 LLM emotions) and V-A coordinates are category-level approximations, these results are presented as global-structure evidence rather than primary proof. The defensible claim is: *the dominant global axis of the tested neural emotion geometry aligns more strongly with an externally defined arousal reference than with a valence reference, while the main ambiguity mechanism is supported independently by centroid margin and classifier uncertainty.*

6 Discussion

The shared geometric principle. Both LLM embedding spaces and the tested brain fMRI dataset exhibit the same general relationship: geometric positioning relative to class prototypes predicts representational certainty. In LLMs this operates via centroid margin predicting logit margin; in the brain, via centroid margin predicting classifier uncertainty. These are complementary implementations of a geometric competition principle. The mechanistic interpretation likely lies in the mathematics of distributed high-dimensional representations: any system encoding discrete categories via distributed activation will produce centroids surrounded by distributed instances, generating a void, a belt, and an ambiguity gradient.

Statistical hierarchy of evidence. The findings differ in evidentiary strength. The *strongest* evidence is the margin-uncertainty coupling: brain centroid margin predicts classifier uncertainty (AUC $= 0.81$ vs. 0.56 for raw distance), collapsing under feature shuffling; LLM centroid margin predicts logit margin ($r = 0.99$, permutation $p = 0.00$). *Strong* evidence: the valence–arousal axis inversion, supported by independent permutation tests on PC-axis correlations ($p < 0.05$ for all three systems). *Supportive* evidence: the 11D RDM inversion, whose bootstrap CI for fine-tuned LLMs is fully negative. *Descriptive evidence only*: the 48D RDM magnitude (3-point Pearson, wide CI) and the void–belt density similarity (Wasserstein permutation $p = 0.46$). The primary evidence for global inversion is therefore the valence–arousal axis analysis, not the 3-emotion RSA magnitude.

The relational paradox: different priorities, not failure. The near-opposite RDM directionality does not indicate that either system is wrong. LLMs trained on text corpora encode emotion co-occurrence statistics from natural language — "fear" and "sadness" share negative hedonic contexts and appear in similar syntactic environments, placing their centroids near each other along the primary valence axis. The brain evolved to respond to environmental threats and rewards: fear involves a physiological arousal response fundamentally different from the low-arousal withdrawal of sadness, and separating states by activation intensity is most predictive of appropriate behavioural response. Neither system is incorrect; they represent emotion faithfully within their respective operational domains.

The low-rank emotion subspace. The 2.6%-capacity finding has direct implications for efficient emotion representation. Accuracy in 20D equals or exceeds 768D in all four tested variants. For real-time affective computing or edge deployment, projecting to the emotion-discriminative subspace is sufficient; larger representations may be counterproductive without fine-tuning.

Limitations. Our evidence covers one text emotion dataset (6 classes) and one brain fMRI dataset (5 emotions, 40 subjects, one ROI atlas). The 3-emotion cross-system label mapping (fear≈afraid, happiness≈delighted, sadness≈depressed) relies on conceptual equivalence and introduces non-trivial uncertainty into all RSA comparisons. The V-A analysis relies on externally assigned category-level coordinates rather than participant-specific ratings; the global-axis findings should be interpreted as supportive evidence, not as the main proof of the margin-based ambiguity mechanism. All findings are observational; no causal evidence is established.

7 Conclusion

We have presented five findings characterising the geometry of emotion representations in LLMs and human fMRI. The core result is a paradox: a geometric competition principle for representational ambiguity is observable across both systems — LLMs via centroid margin predicting logit margin; the brain via centroid margin predicting classifier uncertainty — yet global representational organisation is inverted: LLMs sort emotions by valence, the brain by arousal.

Three findings have immediate practical value. The low-rank emotion subspace (2.6% of embedding capacity, 20 of 768 dimensions) shows that efficient emotion encoding does not require large representations. The margin-vs-distance contrast (AUC 0.81 vs. 0.56) provides a reusable diagnostic distinguishing relational from radial geometric structure. The V-A inversion suggests that training LLMs to distinguish arousal levels within the same valence category may be necessary for biological alignment — a hypothesis for future experimental work.

More broadly, the coexistence of shared local geometry with globally inverted structure suggests that cross-system alignment in affective representation is not a binary question: two systems can share the same local competitive logic while differing fundamentally in what they prioritise globally.

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A V-A Reference Coordinate Robustness

The external V-A reference coordinates used in Section 5.5 are category-level approximations from the circumplex literature [Posner et al., 2005]. To confirm that the axis-dominance results are not artefacts of these specific coordinate assignments, we ran a permutation test ($n = 1000$) for each system.

Procedure. For each of the three systems (fine-tuned LLM, pretrained LLM, brain fMRI), emotion centroids are projected into the leading 2D PCA subspace. Valence and arousal correlations are computed with the reference coordinates (Table 1). A null distribution is constructed by shuffling the V-A label assignments across emotions 1,000 times; each shuffle recomputes the maximum of the valence and arousal correlations. The permutation p-value is the proportion of shuffles that equal or exceed the observed maximum correlation.

Results.

System	Dominant axis	Observed r	Permutation p ($n = 1000$)
Fine-tuned LLM	Valence	0.97	0.009
Pretrained LLM	Valence	0.97	0.013
Brain fMRI	Arousal	0.96	0.033

All three systems reach conventional significance under label permutation. The axis assignments — valence-dominant for both LLM types, arousal-dominant for the brain — are not recoverable by random reassignment of the reference coordinates. The qualitative inversion is robust to the specific coordinate values used.

B Phase 5 Full Results

Phase 5 evaluates only the two fine-tuned final-layer models against the 4,038-sample validation set.

Metric	BGE-FT-Final	MPNet-FT-Final
Total samples	4,038	4,038
Overlap samples	≈76 (1.88%)	≈71 (1.76%)
Dist-logit Pearson r	0.9572	0.9884
Logit agreement in overlap	93.4%	100%
Dominant overlap pair	Happin./Love	Happin./Love

The 100% logit agreement for MPNet-FT-Final means every sample geometrically closer to a competing centroid also has a higher raw logit score for that competing class. Embedding geometry is the direct mechanistic basis for classifier confidence in this model.

C Overlap Metrics: All Final-Layer Variants

Variant	Overlap (%)	Density Peak	Certainty Buffer	Void Onset
MPNet-FT-Final	1.76	0.562	+1.875	≈0.4375
BGE-FT-Final	1.88	0.688	+1.750	≈0.4375
MPNet-Base-Final	18.45	0.938	0.000	≈0.8125
BGE-Base-Final	19.22	0.938	+0.125	≈0.6875
BGE-Base-Mid	28.14	—	—	—
MPNet-Base-Mid	16.92	—	—	—

D fMRI Sample Size Justification

Dataset	N subjects	Venue
Individual Brain Charting (IBC)	12	Nature Scientific Data
Precision Functional Mapping (Children)	12	Dosenbach Lab / PMC
Natural Scenes Dataset (NSD)	8	NSD Website
This study (DS005700)	**40**	OpenNeuro

With $N = 40$, LOSO cross-validation leaves 39 training subjects per fold — well within the range where logistic regression on 48 features is well-determined. The 5-class chance level is 0.20; our LOSO accuracy of 0.56 (95% CI: 0.49–0.63, permutation $p < 0.001$) represents an absolute lift of 0.36 above chance.

E 48D→11D Transformation: ROI-to-Network Mapping

Anatomical lobe features (5): Mean activation within (1) frontal cortex ROIs, (2) temporal cortex ROIs, (3) parietal cortex ROIs, (4) occipital cortex ROIs, (5) cingulate/limbic ROIs.

Functional network features (5): Mean activation within (6) Default Mode Network ROIs [Li et al., 2014] — bilateral medial PFC, posterior cingulate, angular gyrus; (7) Salience Network ROIs [Seeley, 2019] — bilateral anterior insula, anterior cingulate cortex; (8) Central Executive Network ROIs [Miller and Cohen, 2001] — bilateral dorsolateral PFC, lateral parietal cortex; (9) visual processing network ROIs — bilateral occipital and fusiform regions; (10) somatomotor network ROIs — bilateral pre/postcentral gyrus.

Neighbour context feature (1): Mean absolute difference between adjacent ROI activations within each anatomical region, capturing local inter-regional coordination.

F Image Experiments: Negative Result

An extension to 120,000 affective image embeddings (CLIP-based) was explored. Image embeddings did not yield clear cluster separation: silhouette scores remained near zero and the void-belt density structure was absent. The failure mode was insufficient cluster separation across emotion categories. This negative result informs the scope of the geometric competition principle: it is observable in text-based LLM embeddings and brain fMRI representations under the systems and datasets tested, but was not apparent in CLIP image embeddings. The modality (text vs. image), CLIP's training objective, and the emotion taxonomy used are candidate explanations.

G Compute and Reproducibility

Hardware. LLM embedding extraction: HPC cluster (GPU nodes). Brain analysis: standard compute (CPU-bound statistical analyses).

Code and data. Code and data are provided as supplementary material. The repository contains: (1) embedding extraction scripts for all 8 LLM variants, (2) all six brain analysis scripts, (3) the 48D→11D transformation code, (4) the cross-system global behaviour comparison script, (5) the V-A alignment code, (6) `valence_arousal_reference_table.csv`, (7) requirements and environment specification.

Ethics statement. This study uses only publicly available, de-identified neuroimaging data from the Neural MO dataset, OpenNeuro accession DS005700. No new participants were recruited. Informed consent and ethical approval were obtained by the original dataset creators; this re-analysis does not require additional ethical review.